

REMARKS

The paper is in response to the Final Office Action mailed February 27, 2017 (the “Office Action”). The foregoing section “AMENDMENTS TO THE CLAIMS” defines the status of each claim.

Applicants respectfully request reconsideration of the application in view of the above amendments to the claims and the following remarks. For Examiner’s convenience and reference, Applicants present remarks in the order that the Office Action raises the corresponding issues.

In connection with the prosecution of this case and any related cases, Applicants have, and/or may, discuss various aspects of the disclosure of the cited references as those references are then understood by Applicants. Because such discussion could reflect an incomplete or incorrect understanding of one or more of the references, the position of Applicants with respect to a reference is not necessarily fixed or irrevocable. Applicants thus hereby reserve the right, both during and after prosecution of this case, to modify the views expressed with regard to any reference.

Please note that Applicants do not intend the following remarks to be an exhaustive enumeration of the distinctions between any cited references and the claims. Rather, Applicants present the distinctions below solely by way of example to illustrate some of the differences between the claims and the cited references. Finally, Applicants request that Examiner carefully review any references discussed below to ensure that Applicants’ understanding and discussion of any reference is consistent with Examiner’s understanding.

Unless otherwise explicitly stated, the term “Applicants” is used herein generically and may refer to a single inventor, a set of inventors, an appropriate assignee, or any other entity or person with authority to prosecute this application.

Rejections Under 35 U.S.C. § 102

The Office Action rejects claims 1-10 and 12-15 under 35 U.S.C. § 102(b) over *Mudaliar* (“Avarai Nei” from *Athmarakshaamirtham*, Pub: Ilakkana Achagam, chennai (1897), pp. 536-

537. Retrieved from: TKDL on 9/8/2016). The Office Action also rejects claims 1, 2 and 4-18 under 35 U.S.C. § 102(b) over *Ida et al.* (JP2005145948). According to MPEP § 2131, a claim is anticipated under 35 U.S.C. § 102(a), (b), or (e) “only if each and every element as set forth in the claim is found, either expressly or inherently described, in a single prior art reference.” The reference must show the identical invention in as complete detail as is contained in the claim. Finally, the elements must be arranged or combined as required by the claim.

Applicants do not necessarily agree with Examiner’s statements regarding at least (1) whether Mudaliar or Ida discloses each and every element recited in the claims, (2) whether Mudaliar or Ida teaches the combination presented in the claims, and (3) whether Mudaliar or Ida discloses or inherently includes the features that the Office Action asserts that Mudaliar or Ida includes. Nevertheless, in the interest of expediting prosecution, Applicants have herein amended independent claim 1 to recite subject matter that is not taught or suggested by Mudaliar or Ida. The originally filed and pending claims provide support for the claim amendments.

The Applicant has amended claim 1 and cancelled claims 5,12,13,14 and 15 to recite subject matter that is not taught or suggested by Mudaliar or Ida. Further, the Applicant has amended “composition comprising” into “composition consisting essentially of” to narrow the claim language. Now, Mudalair and Ida are excluded because their compositions necessarily include many more essential ingredients than those of the present claims. In fact, neither Mudalair nor Ida teaches or suggests that a palm leaf extract alone as the only effective anti-diabetic substance in a composition can be used for treating a diabetic subject. Now, by reciting “composition consisting essentially of a substance having an anti-diabetic property, the substance being palm leaf extract” the claims are limited to compositions much different from those in Mudalair and Ida because the compositions in Mudalair and Ida require much more than just “*a substance having an anti-diabetic property, the substance being palm leaf extract which includes biophenols, proteins, lipids, saccharides, minerals and small peptides.*” The amended claim 1 now reads as:

A method of treating a diabetic subject, the method comprising:

- i. providing a composition consisting essentially of a substance having an anti-diabetic property, the substance being palm leaf extract which includes biophenols, proteins, lipids, saccharides, minerals and small peptides; and
- ii. administering the composition in a pharmaceutically effective amount to the diabetic subject to treat diabetes disease.

The Applicant noted that Mudaliar teaches a composition with 31 substances as ingredients which includes *cocos nucifera* (coconut palm) as one of the ingredient of the “Avarai Nei” composition. However, Mudaliar does not disclose which one or more substances is/are used for treating the “all types of Diabetes mellitus” (see page 2 to 4 of Mudaliar). Whereas, claim 1 of the present invention clearly indicates that only one ingredient i.e. the palm leaf extract is needed to treat diabetes disease (see lines 5-8, page 1 of the originally filed specification).

In addition, the amended claim 1 indicates that the palm leaf extract includes biophenols, proteins, lipids, saccharides, minerals and small peptides (see lines 19 -23, page 7 of originally filed specification). These substances are crucial features of the present invention as they contribute anti-diabetic and organ health promoting properties (see lines 25-27, page 8 of the originally filed specification). However, Mudaliar does not disclose the above mentioned substances, therefore it is apparent that teachings of Mudaliar are distinctive over present invention.

Further, the amended claim 1 of the present invention includes the transition phrase “consisting essentially of” with respect to the *substance having an anti-diabetic property, the substance being palm leaf extract*. In view of the above and disclosure of Mudaliar, we can deduce that, the 31 substances are an essential feature of Mudaliar’s composition. However, these components are not needed and excluded from the composition of the present invention by presence of the transitional phrase “consisting essentially of” in claim 1. As such, it is clear that the composition of palm leaf extract of the present invention is different compared to the composition of Mudaliar.

On the separate note, the Examiner has indicated that the Applicant's argument is not found persuasive because Mudaliar discloses that the composition comprising water extract of *Cocos Nucifera* leaf is effective at treating diabetes. It appears that *Cocos Nucifera* in combination with other substances is effective at treating diabetes (see page 4 of Mudaliar) but nowhere, in Mudaliar it is disclosed that *Cocos Nucifera* alone is sufficient to treat diabetes. Therefore, it is not evident to a person skilled in art to specifically choose *Cocos Nucifera* out of 31 substances to treat diabetes. As such, the present invention is distinctive as it teaches that palm leaf extract alone is sufficient to treat diabetes disease (see lines 3-9, page 4 of the originally filed specification). Moreover, the new claims recite subject matter that is not taught or suggested by Mudaliar.

Additionally, the Examiner has indicated that Ida teaches a method of treating diabetes comprising administering a methanol or ethanol extract of *Phoenix rupicola* leaves and it does not require any additional ingredients. Further, Ida teaches that the formulation can be administered as a tablet, capsule, suppository, injections, emulsions or liquid solution with the dose of 50mg/kg. However, it is clearly stated in Ida that there are 4 compounds (compound 1 is cell death inducer and compounds 2, 3 and 4 are antitumor agent) in the invention thereof and specifically compound 4 is extracted from *Phoenix rupicola* (see lines 0018 and 0022, page 4 of Ida et al.). As such, *Phoenix rupicola* is taught to be an antitumor agent, but not an antidiabetic agent, and nowhere does Ida teach that compounds 1, 2, 3, or 4 or extract of *Phoenix rupicola* can be used to treat diabetes. Meanwhile, the presently claimed invention does not include the other specific compounds of Ida and presently claimed invention focuses only on palm leaf extract (see lines 25-27, page 7 of originally filed specification).

Paragraph 23 as cited in the Office Action states that the specifically pure compounds 1, 2, 3, or 4 (e.g., compound (1) (4)) can be obtained by extraction method. Then paragraphs 24-25 teach that purification must be performed, and thereby after such purification to obtain the compounds 1, 2, 3, or 4, there is no longer an extract, especially not one with the substances required in claim 1. Nowhere does it teach a palm leaf extract is obtained that can be used to treat diabetes. Ida focuses on compounds to cause cell death, but not a palm leaf extract that be used to treat diabetes.

A careful reading of paragraph 37 never teaches which compound 1-4 can be used to treat diabetes, and in fact, no compound or extract is taught to be useful for treating diabetes. The paragraph merely states the cell death inducer, which means one of compounds 1-4 may be applied to other diseases, where diabetes is listed, but the paragraph never actually teaches it is useful. This paragraph is more of the “maybe try one of the compounds on other diseases.” There is never an affirmative statement that any of the compounds 1-4 actually can be used to treat diabetes, there is no evidence they can be used to treat diabetes, and no suggestion that the extract of palm leaf can be used to treat diabetes and there is no evidence that a cell death inducer would be actually useful for treating diabetes.

Applicants respectfully point out that pancreatic beta cell death causes diabetes, and thereby nobody of ordinary skill in the art would actually try to treat diabetes by killing pancreatic beta cells.

Applicants also note that the examples of Ida clearly require purification and thereby the resulting compounds are not considered extracts. Further, there is clearly no teaching in Ida that an extract of palm leaf that includes biophenols, proteins, lipids, saccharides, minerals and small peptides can be used to treat diabetes.

In addition, the amended claim 1 clearly indicates that the palm leaf extract includes biophenols, proteins, lipids, saccharides, minerals and small peptides (see lines 19 -23, page 7 of originally filed specification). These substances are crucial features of the present invention as it contributes anti diabetic and organ health promoting properties (see lines 25-27, page 8 of the originally filed specification). Ida does not disclose or teach that the above mentioned substances are included in a pal leaf extract, therefore the presently claimed invention is novel and non-obvious over Ida.

Further, the amended claim 1 of the present invention includes the transition phrase “consisting essentially of” with respect to the *substance having an anti-diabetic property, the substance being palm leaf extract*. In view of the above and disclosure of Ida, we can deduce that, specifically compound 4 from the extract of *Phoenix rupicola* is an essential feature of the Ida composition. However, these components are not needed and excluded from the composition of the present invention by presence of the transitional phrase “consisting essentially of” in claim

1. As such, it is apparent that the composition of palm leaf extract of the present invention is different than the composition of Ida.

The amendments presented herein, taken as a whole, render the rejections moot. The Applicant notes that portions of the claims, other than those specifically amended and/or mentioned above, may contribute to patentability. In light of the references as now understood by the Applicant and the amendments made herein, the claims appear to be allowable. The Applicant therefore respectfully requests the Examiner to withdraw the rejections under 35 U.S.C. § 102 of the claims.

Rejections under 35 U.S.C. § 103(a)

The Office Action rejects claims 1-15, 18 and 19 under 35 U.S.C. § 103(a) over *Mudaliar* in view of *Nursing times* (“The administration of medicines”. Internet publication date: 2007-11-19 [Retrieved from the Internet on: 2016-09-09]. Retrieved from the Internet: <URL: <https://www.nursingtimes.net/clinical-archive/medicine-management/theadministration-of-medicines/288560/article>>). The Office Action also rejects claims 1-20 under 35 U.S.C. § 103(a) over *Ida* in view of *Agasthiyar* (“Nilapanai Kirutham” from Agasthiyar Vaithya Rathina Churukkam, Ed. Pub: Thamarai Noolagam, Chennai (1998, 2nd edition), pp. 69-70. Retrieved from: TKDL on 2/18/2017) and *Nursing times*.

Applicants do not necessarily agree with Examiner’s statements regarding at least (1) whether the cited reference(s) disclose each and every element recited in the claims, (2) whether the cited reference(s) teach the combination presented in the claims, (3) whether the cited reference(s) disclose or inherently include the features that the Office Action asserts that the reference(s) include, and (4) whether the reference(s) or any other source provide a reason to combine the teachings of the reference(s). Nevertheless, in the interest of expediting prosecution, Applicants have herein amended independent claims to recite subject matter that is not taught by any of the cited references, alone or in combination.

Applicants submit that Nursing Times does not cure the deficiencies of *Mudaliar* and/or *Ida* recited herein. Applicants submit that *Agasthiyar* and *Nursing Times* combined do not cure

the deficiencies of Mudaliar and/or Ida recited herein. Thus, the claims are allowable over these combinations of references.

The Applicant has herein amended claim 1 to recite subject matter that is not taught or suggested in Mudaliar and/or Nursing Times or Ida, Agasthiyar and Nursing Times combined. The Applicant notes that this rejection relies on Mudaliar anticipating the subject matter of claim 1 regarding the palm leaf extract. Amended claim 1 clearly indicates that only one ingredient which is palm leaf extract which includes biophenols, proteins, lipids, saccharides, minerals and small peptides is needed to treat diabetes disease (see lines 19 -23, page 7 of originally filed specification). These substances are crucial features of the present invention as it contributes anti diabetic and organ health promoting properties (see lines 25-27, page 8 of the originally filed specification).

Further, it is clear that Mudaliar teaches about ‘Avarai Nei’ which is a therapeutic composition consists of 31 ingredients (see page 2 of Mudaliar). Meanwhile, Nursing Times teaches about clinical procedure involved in administration of medicine in accordance with the administration method used. Further, it teaches about bioavailability of drug and best administration mode for the drug. However, the present invention is not similar to the prior arts as it teaches about extraction of palm leaf and the nutraceutical benefits of the palm leaf extract specifically for treating diabetes. Further, the present invention discloses that the composition of the palm leaf extract in amount of 50-200 mg/kg needs to be administered daily for 15 days to treat diabetes. However, nowhere in both prior arts is the amount of the extract and duration of the administration disclosed. As such, it is impossible to achieve the present invention’s solution by combining both prior arts. Notably, Nursing Times does not teach to exclude all of the other ingredients except for palm leaf extract, and thereby the combination of references does not teach or suggest the present claims.

In addition, the amended claim 1 of the present invention includes the transition phrase “consisting essentially of”. In view of the above and disclosure of Mudaliar and Nursing times, we can deduce that, the 31 components and the method of administration of medicine is an essential feature of the prior arts. However, these components are not needed and excluded from the composition of the present invention by presence of the transitional phrase “consisting

essentially of” in claim 1. As such, it is evident that the composition of the palm leaf extract of the present invention is different and it is not possible to arrive at present invention’s solution even though both prior arts are combined.

In addition, the Nursing Times does not teach or suggest extraction of palm leaf and the nutraceutical benefits of the palm leaf extract for treating diabetes. As such, the combination of Mudaliar and Nursing Times does not teach or suggest the present claims for the same reasons Mudaliar does not teach or suggest the present claims. Thus, Nursing Times does not cure the deficiencies of Mudaliar. As such, Nursing Times is irrelevant to the present invention as it only discloses the method of medicine administration whereas the present invention is not limited to the method of administration of medicine, as it focuses on use of palm extract for diabetes treatment.

The Examiner rejects claims 1-20 under 35 U.S.C. § 103(a) as unpatentable over Ida in view of *Agasthiyar*, in view of *Nursing times* (“The administration of medicines”. Internet publication date: 2007-11-19).

The Examiner has indicated that Ida teaches a method of treating diabetes comprising administering a methanol or ethanol extract of *Phoenix rupicola* leaves and it does not require any additional ingredients. However, Applicants respectfully submit that this assertion is erroneous and nowhere does such a teaching occur in Ida. Meanwhile, *Agasthiyar* teaches that a composition comprising *Phoenix dactyifera* stem bark in 10.24 g is effective in treating diabetes and Nursing times teaches that topical administration is an effective form for locally administering medication. The Office Action alleges that it would have been obvious to the person skilled in art to manipulate palm leaf extract for treating diabetes using topical administration.

Applicants note that stem bark is not a palm leaf, and thereby the combination is defective. There is no evidence that the extract of stem bark of one species has the same components or effective uses as the purified compounds of the leaf of a different species. Thus, there is no logical combination or motivation to make such a combination.

Applicants note that the combination is defective and based on only impermissible hindsight because there is no logical connection to a teaching in Ida to extract from palm leaf and then purify the extract into a pure compound for use as a cell death inducer and then combine that reference with a completely different concept of extraction from stem bark to treat diabetes. These references do not have a common nexus and Nursing Times does not cure this deficiency. If combined, Ida suggests to purify compounds, which necessarily results in something other than the extract of palm leaf that is presently claimed. Thus, the combination is based on impermissible hindsight in an effort to recreate the present invention using the present claims as a roadmap.

In view of above, the Applicant has herein provided arguments as stated below that the present invention is patentable in view of above the objection because the cited prior arts do not teach the present invention. The present invention teaches a palm leaf extract in a pharmaceutically effective amount which is about 100 mg/kg for treating diabetes disease (see lines 12-14, page 10 of originally filed specification). However, none of the above indicated prior arts disclose that palm leaf extract alone could be used for treating diabetes diseases. Further, although Agasthiyar discloses 40 ingredients in combination for treatment of anti-diarrhea and semen retaining, the use of *Phoenix dactyifera* stem bark in 10.24 g in view of Ida is not obvious to the person skilled in the art. This is due to the fact that Ida discloses 4 types of components which have different effects, wherein compound 1 is cell death inducer and compounds 2, 3 and 4 are antitumor agent (see lines 0018 and 0022, page 4 of Ida). Therefore, a skilled artisan will be unable to specifically identify the use of palm leaf extract to treat diabetes disease and it is not possible to arrive to present invention.

Further, as mentioned earlier the amended claim 1 of the present invention includes the transition phrase “consisting essentially of”. In view of the above and disclosure of Ida, Agasthiyar and Nursing times, we can deduce that, the 4 components of Ida, the 40 components of Agasthiyar and the method of administration of medicine are an essential feature of the prior arts. However, these components are not needed and excluded from the composition of the present invention by presence of the transitional phrase “consisting essentially of” in claim 1. As such, it is apparent that the composition of the palm leaf extract of the present invention is

different and it is not possible to arrive at present invention solution even though the prior arts are combined.

As such, it is clear that the cited documents do not disclose or render obvious each and every limitation of the amended claims. Thus, the novel and inventive feature of the present invention lies in overcoming the demerits of the prior arts, and in no way can the present invention be considered as anticipated or obvious since the invention achieves results which are clearly not possible even if one was to get motivated by the cited prior arts.

Therefore, the amendments presented herein, taken as a whole, render the rejections moot. The Applicant noted that portions of the claims other than those specifically amended and/or mentioned above may contribute to the patentability. In light of the references as now understood by the Applicant and the amendments made herein, the claims appear to be allowable. Therefore, the Applicant respectfully requests the Examiner to withdraw the rejections.

CONCLUSION

In view of the foregoing, Applicants submit that the pending claims are allowable. In the event that Examiner finds remaining impediment to a prompt allowance of this application that may be clarified through a telephone interview or overcome by an Examiner's Amendment, Examiner is requested to contact the undersigned attorney.

Dated this 10th day of May, 2017.

Respectfully submitted,

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AMENDMENTS TO THE CLAIMS

This listing of claims replaces all prior versions and listings of claims in the application:

1. **(Currently Amended)** A method of treating a diabetic subject, the method comprising:
 - i. providing a composition comprising consisting essentially of a substance having an anti-diabetic property, the substance being palm leaf extract which includes biophenols, proteins, lipids, saccharides, minerals and small peptides; and
 - ii. administering the ~~said~~ composition in a pharmaceutically effective amount to the diabetic subject ~~so as to provide a pharmaceutically effective amount of the palm leaf extract to treat diabetes disease in the diabetic subject.~~
2. **(Currently Amended)** The method as claimed in claim 1, wherein the extract is derived from leaves of a palm tree species[[],] .
3. **(Previously Presented)** The method as claimed in claim 2, wherein the palm tree species is selected from a group comprising of *Elaeis guineensis*, *Elaeis oleifera*, *Phoenix dactylifera*, *Cocos nucifera* or a combination thereof.
4. **(Original)** The method as claimed in claim 1, wherein the extract is an aqueous, an aqueous alcoholic or a polar extract.
5. **(Cancelled)**
6. **(Original)** The method as claimed in claim 1, comprising administering a pharmaceutically effective amount of the palm leaf extract sufficient to reduce blood sugar in the diabetic subject.

7. **(Original)** The method as claimed in claim 1, comprising administering a pharmaceutically effective amount of the palm leaf extract sufficient to inhibit diabetic oxidative stress in the diabetic subject.
8. **(Original)** The method as claimed in claim 1, comprising administering a pharmaceutically effective amount of the palm leaf extract sufficient to inhibit diabetic complications in the diabetic subject.
9. **(Original)** The method as claimed in claim 1, comprising administering a pharmaceutically effective amount of the palm leaf extract sufficient to inhibit tissue degeneration, cardiovascular disease, kidney dysfunction, loss of cognitive function, weight loss, visual impairment, and vasomotor symptoms in the diabetic subject.
10. **(Currently Amended)** The method as claimed in claim 1, wherein the composition of the palm leaf extract is incorporated in an edible or a topical product[[]] .
11. **(Original)** The method as claimed in claim 10, wherein the edible or topical product comprises of capsules, ointments, lotions and tablets.
- 12.-15. **(Cancelled)**
16. **(Previously Presented)** The method as claimed in claim 1, wherein the palm leaf extract is an alcoholic extract.
17. **(Previously Presented)** The method as claimed in claim 1, wherein the palm leaf extract is an ethanol extract.
18. **(Previously Presented)** The method as claimed in claim 1, wherein the pharmaceutically effective amount is 50-200 mg/kg.

19. **(Previously Presented)** The method as claimed in claim 18, wherein the pharmaceutically effective amount is administered daily for at least 15 days.
20. **(Cancelled)**